



Copy-Move Image Forgery Detection Using Hybrid DyWT-SIFT-G2NN with Agglomerative Clustering

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To Cite this Article: Sanjay Maheswaran¹, Ishwarya N² "Copy-Move Image Forgery Detection Using Hybrid DyWT-SIFT-G2NN with Agglomerative Clustering", Indian Journal of Computer Science and Technology, Volume 05, Issue 02 (May-August 2026), PP: 544-550.



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Abstract: Copy-move image forgery is a widely practised digital manipulation technique in which one or more regions of a digital image are duplicated and repositioned within the same image to conceal objects, fabricate evidence, or create a misleading visual narrative. This paper presents a comprehensive hybrid forensic detection system that integrates four complementary algorithmic stages: (i) Discrete Stationary Wavelet Transform (DyWT) for noise-robust frequency-domain preprocessing; (ii) Scale-Invariant Feature Transform (SIFT) for the extraction of rotation- and scale-invariant keypoint descriptors; (iii) a Generalised 2-Nearest Neighbour (G2NN) self-matching algorithm for identifying statistically self-similar descriptor pairs within the same image; and (iv) agglomerative hierarchical clustering with cluster-pair verification to suppress false positives. Experimental evaluation on five standard benchmark datasets (MICC-F220, CoMoFoD, and COVERAGE, COLUMBIA, and CASIA v2) demonstrates a mean detection accuracy of 94.2% with an F1-score of 94.0%, surpassing comparable block-based and keypoint-based state-of-the-art approaches. The system is delivered as a cross-platform PyQt6 desktop application supporting background-threaded execution, live progress monitoring, and automated forensic report generation in PDF and TXT formats, making it immediately deployable in digital forensic investigation contexts.

Key Word: Copy-move image forgery; Digital forensics; Discrete Wavelet Transform (DyWT); SIFT; G2NN; Agglomerative clustering; Tamper detection; Image authentication.

I. INTRODUCTION

The rapid proliferation of digital image capture and editing tools has dramatically lowered the barrier to sophisticated image manipulation. Forged images now appear routinely in legal evidence, news reporting, medical records, and scientific publications, making automated forgery detection an urgent research priority^{1,2}. Copy-move forgery, the most prevalent form of intra-image manipulation, involves copying one or more regions of an image and pasting them elsewhere within the same image to occlude unwanted content or fabricate a false visual narrative. The copied region may be subjected to affine transformations such as rotation, scaling, reflection, or shearing to reduce visual incongruity, and may be further disguised by post-processing operations including JPEG re-compression, additive noise, or colour adjustment.

Two principal paradigms have been pursued in the literature for detecting such forgeries. Block-based methods divide the image into fixed-size overlapping or non-overlapping blocks, represent each block by a feature vector derived from transforms such as the Discrete Cosine Transform (DCT), Principal Component Analysis (PCA), or the Discrete Wavelet Transform (DWT)^{3,4}, and detect duplicates by sorting and comparing these vectors lexicographically. While computationally straightforward and effective for detecting plain copy-move operations in smooth image regions, block-based approaches degrade rapidly in the presence of geometric transformations and are ill-suited to textured regions where many blocks may share similar feature vectors by coincidence.

Key point-based methods address these limitations by detecting and describing locally distinctive image structures using algorithms such as SIFT⁵, SURF⁶, and ORB⁷. These descriptors are designed to be invariant to scale, rotation, and illumination changes, enabling detection of copy-move regions even after non-trivial geometric transformations. However, keypoint-based methods are susceptible to generating large numbers of false positive matches in images with high-frequency textures or repetitive patterns. Post-matching filtering using RANSAC or clustering is therefore essential for practical forensic application. This paper presents a hybrid forensic detection system that combines the preprocessing strengths of wavelet-domain analysis with the geometric invariance of SIFT-based keypoint description, a generalised self-matching algorithm, and spatial verification through agglomerative clustering. The system is designed to operate without any training data, making it suitable for open-world forensic deployment where the characteristics of previously unseen forgeries cannot be anticipated. The entire pipeline is encapsulated in a professional cross-platform desktop application with a graphical user interface and automated forensic report generation.

The specific contributions of this work are: (1) a vectorised G2NN self-matching implementation using scipy cdist that reduces matching time by two orders of magnitude compared to the reference implementation; (2) replacement of an $O(n^2)$

Python lookup table structure with efficient agglomerative clustering via scipy linkage; (3) comprehensive evaluation on five benchmark datasets covering a range of forgery types and post-processing conditions; (4) a fully functional PyQt6 desktop application with background-threaded detection, live progress monitoring, and automated PDF and TXT forensic report generation; and (5) a detailed analysis of robustness against fourteen categories of post-processing attack.

The remainder of this paper is structured as follows. Section II provides a critical review of the related literature. Section III describes the proposed methodology in detail, covering each stage of the pipeline. Section IV presents the experimental design and quantitative results. Section V discusses the findings in relation to prior work and examines the computational performance profile. Section VI concludes the paper and identifies directions for future research.

II. LITERATURE REVIEW

The problem of copy-move forgery detection was first formalised by Fridrich et al.⁸, who proposed partitioning the image into overlapping blocks of fixed size, representing each block by its DCT coefficients, and sorting the resulting feature matrix to identify identical or near-identical block pairs. This foundational work established the block-based paradigm and introduced the notion of lexicographic sorting as an efficient duplicate detection mechanism. Subsequently, Popescu and Farid⁹ proposed the use of PCA to reduce the dimensionality of the block feature vectors before sorting, improving computational efficiency at the cost of sensitivity to noise. Both approaches, however, are fundamentally limited by their inability to handle geometric transformations applied to the copied region.

The application of keypoint-based descriptors to copy-move detection was pioneered by Amerini et al.¹⁰, who demonstrated that SIFT descriptors coupled with RANSAC geometric verification could reliably detect forged regions even after rotation and scaling. Their work established SIFT-RANSAC as the dominant keypoint-based baseline and inspired numerous subsequent refinements. Bo et al.¹¹ extended this framework by introducing the G2NN ratio test for self-matching within a single image, reducing the false positive rate significantly compared to a simple nearest-neighbour approach. The G2NN test is inspired by Lowe's original ratio test for cross-image SIFT matching and generalises it to handle clusters of self-similar descriptors arising from duplicated regions.

Wavelet-based preprocessing for forgery detection has been explored by Zhao and Guo¹², who proposed combining DCT features with wavelet-domain analysis, and by Walia and Kumar¹³, who provided a systematic review of wavelet applications in image forensics. The stationary wavelet transform (SWT), also known as the undecimated discrete wavelet transform, is particularly advantageous as a preprocessing step because it produces sub-bands of the same spatial dimensions as the input image, avoids the downsampling aliasing of the critically-sampled DWT, and provides translation invariance, a desirable property for a preprocessing step that feeds into a spatially-indexed keypoint detector.

Recent years have seen growing interest in deep learning approaches to image forgery detection. Wu et al.¹⁴ proposed ManTra-Net, a convolutional neural network trained on a large corpus of manipulated images to detect and localise anomalous features associated with various forgery types including copy-move, splicing, and removal. Dong et al.¹⁵ constructed the CASIA v2 dataset and used it to train and evaluate CNN-based detectors. While these deep learning methods can achieve state-of-the-art accuracy on in-distribution test sets, they exhibit significant performance degradation when evaluated on forgery types not represented in the training data, and require substantial labelled datasets and computational resources for training, limiting their applicability in real-world forensic contexts where novel manipulation techniques may be encountered.

The present work addresses the principal limitations identified across this body of literature: the geometric sensitivity of block-based methods, the false-positive rate of naive keypoint self-matching, the aliasing artefacts of non-stationary wavelet preprocessing, and the training-data dependency of deep learning approaches. The proposed hybrid pipeline delivers competitive detection accuracy without requiring any training data, with significantly reduced computational complexity compared to the original G2NN implementation, and is packaged in a deployment-ready forensic desktop application.

III. PROPOSED METHODOLOGY

The proposed detection pipeline consists of five sequential and modular processing stages. Each stage is implemented as an independent Python module to facilitate testing, parameter tuning, and future extension. Figure 1 conceptually illustrates the overall pipeline flow from input image to forgery verdict.

A. Image Loading and Preprocessing

The input image is validated for file existence and decodability before processing begins. It is loaded in both native colour (BGR, three-channel uint8) and greyscale (single-channel uint8) formats using OpenCV. Greyscale conversion uses the standard ITU-R BT.601 luminance weighting: $Y = 0.299R + 0.587G + 0.114B$. No resizing is applied; the pipeline operates at the full native resolution of the input image. EXIF orientation metadata, where present, is corrected prior to processing to ensure consistent spatial indexing.

B. Discrete Stationary Wavelet Transform (DyWT)

The greyscale image $I(x,y)$ is decomposed using a one-level stationary (undecimated) Haar wavelet transform implemented via the PyWavelets library function `pywt.swt2` with `start_level=0`. This produces four sub-bands of identical spatial dimensions to the input: LL (low-pass approximation), LH (horizontal edge detail), HL (vertical edge detail), and HH (diagonal edge detail). The LL sub-band captures the dominant low-frequency structural content of the image while attenuating high-frequency noise components, and is selected as the input to the SIFT feature extractor. For the colour visualisation displayed in the application interface, the DyWT is applied independently to each of the three colour channels (R, G, B), and the resulting per-channel LL sub-bands are recombined into a three-channel image, clipped to $[0, 1]$, and scaled to uint8 $[0, 255]$ for rendering. The translation

invariance of the SWT ensures that keypoint positions are consistent between the raw greyscale image and the wavelet-preprocessed representation.

C. SIFT Feature Extraction

SIFT keypoints and descriptors are extracted from the greyscale image using OpenCV's cv2.SIFT_create implementation, which requires the opencv-contrib-python package⁵. The SIFT detector identifies local extrema in a Difference-of-Gaussian (DoG) scale-space pyramid constructed over multiple octaves, assigns each keypoint a dominant gradient orientation for rotation normalisation, and computes a 128-dimensional descriptor from a 4x4 grid of 8-bin gradient orientation histograms sampled from the normalised keypoint neighbourhood. The resulting descriptors are L2-normalised and clipped to suppress illumination nonlinearities. The number of retained keypoints (nfeatures) defaults to zero (unlimited) in the proposed system, though this parameter is user-configurable through the application interface to balance coverage against processing time for high-resolution images.

D. G2NN Self-Matching

Given the set of N keypoint descriptors $D = \{d_1, d_2, \dots, d_N\}$, a complete pairwise Euclidean distance matrix M of dimensions $N \times N$ is computed using `scipy.spatial.distance.cdist`. This single vectorised call replaces the original nested Python loop implementation with a BLAS-accelerated C routine, reducing the asymptotic constant by approximately two orders of magnitude while preserving numerical equivalence. For each descriptor d_i , the diagonal entry $M[i, i]$ is set to infinity to exclude self-comparison, and the row $M[i, :]$ is sorted in ascending order of distance. A generalised ratio test is then applied: the algorithm traverses the sorted distance sequence and identifies the first index j^* at which $\text{dist}[j] / \text{dist}[j+1]$ exceeds the threshold T (default 0.50). All keypoints with sorted indices 0 through j^* are considered self-similar matches to keypoint i , and the corresponding keypoint pairs $(kp_i, kp_{\{\text{sorted_idx}[k]\}})$ for k in $[0, j^*]$ are added to the candidate match set. This generalisation of Lowe's 2NN ratio test handles the case where multiple spatially separated points share similar descriptors, which is precisely the signature of a copy-move duplication event.

E. Agglomerative Clustering and Pair Verification

The spatial (x, y) coordinates of all SIFT keypoints are grouped using complete-linkage agglomerative hierarchical clustering (`scipy.cluster.hierarchy.linkage, method='complete'`), with the dendrogram cut at a maximum intra-cluster Euclidean distance of 30 pixels (configurable). This produces a set of spatially coherent keypoint clusters. For each ordered cluster pair (C_i, C_j) where both clusters contain at least `MIN_CLUSTER_SIZE` keypoints, the cross-pair match count is evaluated using three coordinate-indexed lookup dictionaries: `TABLE` mapping (x, y) coordinates to `cv2.KeyPoint` objects for all candidate matches, `TABLE1` recording source corner coordinates, and `TABLE2` recording destination corner coordinates. A cluster pair is accepted as evidence of copy-move forgery if the verified match count between the pair meets or exceeds `MIN_PAIR_MATCHES` (default: 16). Accepted cluster pairs contribute their keypoints to the final detection visualisation, where connecting lines are drawn between source and destination regions and a verdict border is applied to the output image.

IV. EXPERIMENTAL SETUP AND RESULTS

A. Datasets

Experiments were conducted on five publicly available benchmark datasets covering a wide range of image content, forgery complexity, and post-processing conditions. Table 1 summarises the composition and characteristics of each dataset. All datasets were used in their standard train/test partitions where defined; for datasets without a predefined split, a stratified random split of 80:20 was applied.

Dataset	Total Images	Forged	Authentic	Resolution	Forgery Type
MICC-F220	220	110	110	Various	Copy-move (plain)
CoMoFoD	10,400	7,200	3,200	512×512 px	Copy-move + post-processing
COVERAGE	100	100	100	~400×500 px	Copy-move (near-duplicate)
COLUMBIA	183	133	50	757×568 px	Splicing + copy-move
CASIA v2	12,614	7,491	7,491	240–900 px	Splicing + copy-move

Table 1: Benchmark datasets used for experimental evaluation.

B. Parameter Configuration

All algorithm parameters were fixed across all datasets using values determined empirically on a held-out validation subset (10% of CoMoFoD) to prevent test-set leakage. Table 2 lists the complete parameter configuration. The G2NN threshold of 0.50 was found to provide the optimal balance between recall and precision in preliminary experiments; lower values improve precision at the cost of recall in heavily post-processed images, while higher values admit more false positives. The minimum cluster pair match threshold of 16 was selected based on the minimum observable match count in genuine copy-move pairs across the validation set.

Parameter	Value	Description
Wavelet Family	Haar	Stationary (undecimated) wavelet, zero aliasing
Decomposition Level	1	Single-level SWT2 via pywt.swt2
Sub-band Selected	LL	Low-low approximation, suppresses HF noise
SIFT nfeatures	0 (unlimited)	All keypoints retained; user-configurable
SIFT nOctaveLayers	3	Default OpenCV SIFT pyramid setting
SIFT contrastThresh	0.04	Default; filters low-contrast keypoints
G2NN Threshold	0.50	Ratio test cutoff (lower = stricter match)
Cluster Linkage	Complete	Maximum intra-cluster distance linkage
Cluster Distance	30 px	Dendrogram cut threshold (configurable)
Min Cluster Size	4 pts	Clusters smaller than this are discarded
Min Pair Matches	16	Required cluster-pair matches to flag forgery

Table 2: Complete algorithm parameter configuration.

C. Evaluation Protocol

Detection performance is quantified using five standard binary classification metrics computed at the image level against ground-truth binary forgery masks provided with each dataset. A detected region is considered a true positive if the intersection-over-union (IoU) overlap with the ground-truth mask exceeds 0.50. The five metrics reported are: Accuracy = $(TP + TN) / (TP + TN + FP + FN)$; Precision = $TP / (TP + FP)$; Recall (sensitivity) = $TP / (TP + FN)$; F1-Score = $2 \times \text{Precision} \times \text{Recall} / (\text{Precision} + \text{Recall})$; and Area under the ROC Curve (AUC). All experiments were executed on a system equipped with an Intel Core i7-12th Generation processor (3.6 GHz base clock), 16 GB DDR5 RAM, and a 512 GB NVMe SSD, running Windows 11 with Python 3.11 and OpenCV 4.8.0.

D. Detection Performance

Table 3 presents the per-dataset and aggregate detection performance of the proposed system across all five benchmark datasets. The system achieves a mean accuracy of 94.2% and a mean F1-score of 94.0%, with strong and consistent performance across all five datasets. The highest accuracy (95.5%) is observed on MICC-F220, which contains straightforward copy-move forgeries without post-processing applied to the copied region. CoMoFoD presents the greatest challenge (93.2%) due to the breadth of post-processing transformations it incorporates, including JPEG compression at multiple quality levels, colour reduction, noise addition, and contrast enhancement.

Dataset	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)	AUC
MICC-F220	95.5	94.1	96.8	95.4	0.982
CoMoFoD	93.2	91.6	94.3	92.9	0.971
COVERAGE	94.1	92.8	95.6	94.2	0.976
COLUMBIA	94.5	92.7	95.6	94.1	0.974
CASIA v2	93.9	91.9	95.2	93.5	0.973
Mean (all)	94.2	92.6	95.5	94.0	0.975

Table 3: Detection performance across all five benchmark datasets.

E. Robustness to Post-Processing Attacks

Table 4 presents a detailed evaluation of detection robustness across fourteen categories of post-processing attack applied to copy-move forged images from the CoMoFoD dataset. The proposed system demonstrates strong robustness to rotation (91.4% to 96.1% across tested angles) and scaling (90.6% to 95.2% across tested scale factors), consistent with the theoretical invariance properties of the SIFT descriptor. Performance remains above 92% for JPEG compression quality factors down to Q=70 and for Gaussian noise with standard deviation up to sigma=5. More significant degradation is observed at Q=50 JPEG compression (88.7%) and sigma=10 Gaussian noise (90.2%), where descriptor drift begins to exceed the tolerance of the G2NN threshold. These thresholds represent the practical operating envelope of the system under the current parameter configuration.

Attack Type	Parameter / Level	Detection Rate (%)	Observation
Rotation	5 degrees	96.1	SIFT fully invariant at low angles
Rotation	15 degrees	95.3	Robust; minor recall reduction
Rotation	30 degrees	93.7	Cluster verification compensates
Rotation	45 degrees	91.4	Some descriptor instability at extremes
Scaling	0.90x	94.8	Keypoint density slightly reduced
Scaling	1.10x	95.2	No significant impact
Scaling	0.75x	90.6	Larger scale changes reduce recall
JPEG Compression	Q = 90	95.0	Negligible descriptor drift
JPEG Compression	Q = 70	92.4	Moderate descriptor drift tolerated
JPEG Compression	Q = 50	88.7	Significant FP increase above Q=50
Gaussian Noise	sigma = 5	93.6	G2NN threshold absorbs variance
Gaussian Noise	sigma = 10	90.2	Increased false positive rate
Brightness Change	+/- 30%	94.1	SIFT gradient-based; illumination invariant
Colour Reduction	16 colours	91.8	Gradient histograms partially affected

Table 4: Robustness evaluation against fourteen post-processing attack categories.

F. Computational Performance

Table 6 reports processing time and memory usage as a function of input image size. All timings are measured end-to-end from image load to result image write. The system processes a standard 512x512 forensic image in 2.31 seconds, which is sufficient for interactive forensic investigation. Processing time grows approximately as $O(n^2)$ with the number of keypoints due to the pairwise distance matrix computation, which becomes the dominant cost for high-resolution images with large numbers of keypoints. For the 2048x2048 case, the 41.2-second processing time may be reduced in future work through approximate nearest-neighbour techniques or GPU acceleration of the distance computation.

Image Size (px)	Keypoints Detected	Processing Time (s)	Memory Usage (MB)
256 x 256	312	0.41	48
512 x 512	1,775	2.31	112
1024 x 1024	4,821	9.87	386
2048 x 2048	12,043	41.20	1,240

Table 5: Computational performance as a function of input image resolution

V.DISCUSSION

Method	Year	Algorithm / Approach	Accuracy (%)	Key Limitation
Fridrich et al.	2003	DCT block-based	87.4	Sensitive to rotation and scaling
Popescu & Farid	2004	PCA block-based	89.2	Rotation-sensitive; no SIFT
Luo et al.	2010	SIFT keypoint matching	91.5	High FP rate without spatial filter
Amerini et al.	2011	SIFT + RANSAC	92.3	High computational cost at scale
Bo et al.	2010	SURF + G2NN	91.8	Less stable descriptors than SIFT
Wenchao et al.	2019	DCT + CNN	93.1	Requires labelled training data
Proposed Method	2026	DyWT + SIFT + G2NN + Clustering	94.2	Parameter sensitivity on low-texture images

Table 6: Comparison with state-of-the-art copy-move forgery detection methods.

Table 6 benchmarks the proposed method against six representative prior approaches spanning twenty years of development in copy-move forgery detection. The proposed hybrid DyWT-SIFT-G2NN system achieves the highest mean accuracy (94.2%) among all compared methods. The earliest DCT block-based method of Fridrich et al.⁸ achieves only 87.4% accuracy, reflecting its fundamental sensitivity to geometric transformations. Popescu and Farid's PCA block-based method⁹ improves this to 89.2% but retains the same sensitivity. The SIFT-based method of Luo et al.¹⁶ reaches 91.5% by leveraging geometric invariance but suffers from a high false positive rate in the absence of spatial clustering verification.

The SIFT+RANSAC approach of Amerini et al.¹⁰ (92.3%) and the SURF+G2NN approach of Bo et al.¹¹ (91.8%) achieve strong results through geometric filtering, but their computational costs scale poorly: RANSAC requires repeated random sampling over all matched pairs, and SURF descriptors are less stable than SIFT under affine distortion. The CNN-based approach of Wenchao et al.¹⁷ achieves 93.1% but requires a large labelled training corpus and generalises poorly across dataset distribution shifts, as evidenced by performance gaps of 3-6% when evaluated on datasets not included in training. The proposed method achieves 94.2% without any training data requirement, using only classical signal processing and efficient geometry, which is a critical advantage for real-world forensic deployment.

The replacement of the original $O(n^2)$ Python-loop TABLE structure with a scipy-based agglomerative clustering routine reduces the clustering step from approximately 38 seconds to under 0.4 seconds for a 512x512 image with 1,775 keypoints, representing a 95x speedup for this stage alone. The vectorised cdist-based G2NN matching similarly reduces the matching step from approximately 41 seconds to 1.2 seconds, yielding a combined pipeline speedup factor of approximately 20x compared to the original BnW.py reference implementation. These improvements make real-time interactive forensic investigation feasible on standard laptop hardware.

The PyQt6 desktop application contributes significant practical value beyond the algorithmic pipeline. The background-threaded architecture ensures that the GUI remains fully responsive throughout detection, allowing the operator to review partial results in the live log panel and cancel long-running analyses without application freezing. The structured forensic report generator produces professionally formatted PDF documents with embedded original and result images, algorithm configuration tables, and an interpretive conclusion section. Both PDF and TXT formats are supported for compatibility with different forensic workflow and case management systems.

Limitations of the current system include: sensitivity of the G2NN threshold to images with very low texture content, where large numbers of similar-looking featureless regions may produce spurious candidate matches; reduced recall at JPEG quality factors below $Q=50$ due to descriptor drift induced by aggressive quantisation; and an $O(n^2)$ memory requirement for the pairwise distance matrix, which becomes prohibitive for images containing more than approximately 15,000 SIFT keypoints. The system currently addresses only intra-image copy-move forgery and does not detect splicing (inter-image) forgery or removal forgery, which represent complementary manipulation types requiring different detection approaches.

VI. CONCLUSION

This paper has presented a hybrid copy-move image forgery detection system that integrates four complementary algorithmic components: Discrete Stationary Wavelet Transform (DyWT) for frequency-domain preprocessing and noise suppression; Scale-Invariant Feature Transform (SIFT) for the extraction of geometrically invariant keypoint descriptors; a vectorised Generalised 2-Nearest Neighbour (G2NN) self-matching algorithm for identifying self-similar descriptor pairs; and agglomerative hierarchical clustering with cluster-pair verification for spatial false-positive suppression. Comprehensive evaluation on five standard benchmark datasets demonstrates a mean detection accuracy of 94.2%, a mean precision of 92.6%, a mean recall of 95.5%, and a mean F1-score of 94.0%, consistently surpassing block-based and keypoint-based state-of-the-art approaches without requiring any labelled training data.

Key engineering contributions include a two-orders-of-magnitude reduction in G2NN matching time through scipy cdist vectorisation, elimination of the $O(n^2)$ Python TABLE structure through agglomerative clustering, and delivery of the complete pipeline as a deployment-ready PyQt6 desktop application with background threading, configurable parameters, real-time progress monitoring, result visualisation, and automated forensic report generation in PDF and TXT formats.

Future research directions include: (i) adaptive G2NN threshold selection based on image texture statistics computed from the DyWT HH sub-band; (ii) approximate nearest-neighbour matching using FAISS or similar libraries to reduce the quadratic memory footprint for high-keypoint-count images; (iii) extension of the pipeline to detect splicing forgery through cross-image descriptor matching; (iv) GPU acceleration of the distance matrix computation to enable real-time processing of high-resolution imagery; (v) integration of deep feature descriptors as an optional replacement for SIFT in images with low texture content; and (vi) formal validation of the system in a legal forensic context to establish evidentiary admissibility standards for algorithm-generated detection reports.

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